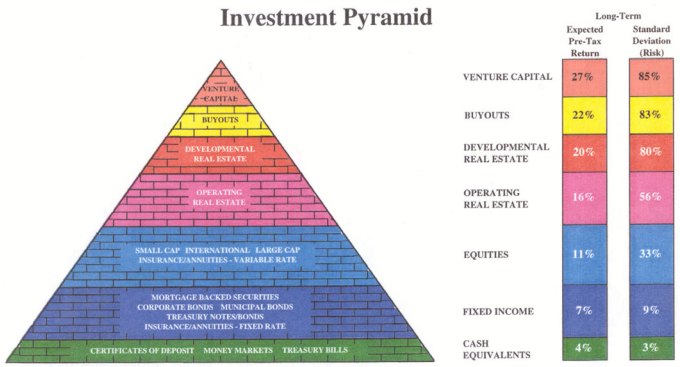
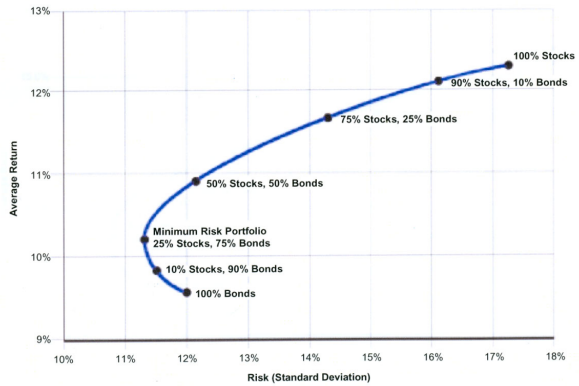
EAS5450
Engineering Entrepreneurship I
Apr 29, 2025

FINANCIAL FORESIGHT
- PERSONAL -

Penn Engineering

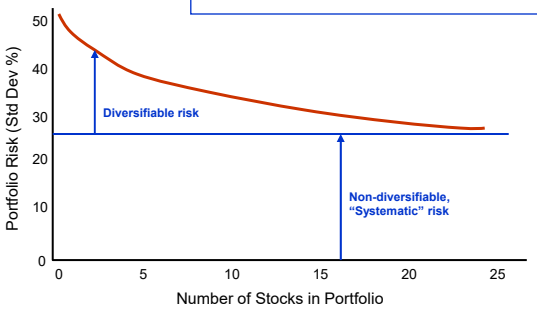




Long-Term NASDAQ Common Stock Performance

NASDAQ Composite Average: Yahoo! Finance.com 4/28/2025

- **Systematic Risk**
  - Risk associated with macroeconomic conditions. Diversification cannot mitigate this risk
- **Diversifiable Risk**
  - Risk associated with the individual firm that can be mitigated through diversification



Stock Risk Factors

- Non-diversifiable, "Systematic" Risk Factors
  - Congress votes a massive tax cut
  - Federal Reserve increases interest rates
  - Economy enters a recession
  - Republicans lose control of Congress
  - Oil-producing countries reduce output: prices rise
- Diversifiable, "Unsystematic" Risk Factors
  - Raw material supplier goes bankrupt
  - Low-cost competitor enters company's market
  - Breakthrough discovery in company lab
  - Fraudulent accounting scheme uncovered
  - COGS very sensitive to oil prices

β

A company's "beta" measures the volatility ("riskiness") of a company's common stock price relative to the volatility of a fully diversified portfolio of common stocks.

- $\beta=0$: Risk-free stock (hypothetical)
- $\beta<1$: Company less risky than diversified portfolio
- $\beta=1$: Company risk equivalent to diversified portfolio
- $\beta>1$: Company more risky than diversified portfolio

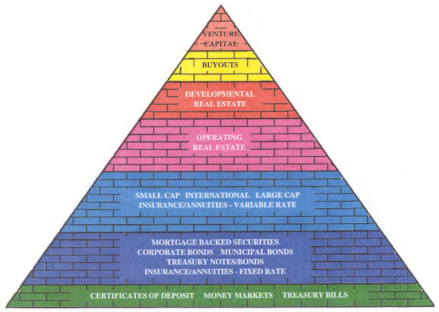
Industry & Company Betas

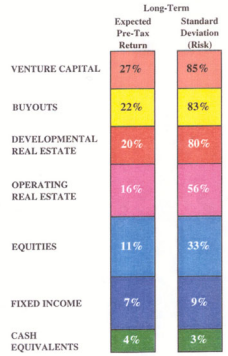
Source: Standard and Poor's Compustat.

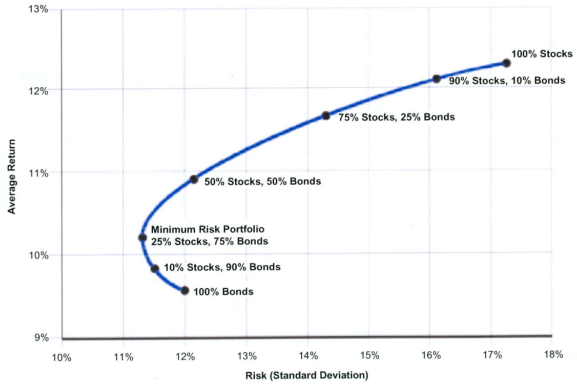
Industry Median Equity Betas		Company Betas	
Industry*	**Beta**	**Company**	**Beta**
Air transportation (20)	1.24	Advanced Micro Devices	2.06
Aircraft (3)	0.58	Aetna	0.69
Apparel and other finished products (13)	0.98	Alaska Air	1.04
Commercial banks (30)	0.74	Amerada Hess	0.53
Communications equipment (24)	1.83	Apple Computer	1.55
Computers and office equipment (3)	1.29	Bank of America	0.97
Electronic components, accessories (4)	1.22	Boeing	0.82
Food and related products (9)	0.37	Burlington Resources	0.66
Gold and silver ores (5)	0.45	ChevronTexaco	0.62
Health services (22)	0.61	Computer Associates	1.84
Motion pictures (9)	0.55	Deere	0.57
Motor vehicle equipment (37)	0.89	Dell Computer	2.14
Petroleum refining (35)	0.67	Hewlett Packard	1.45
Chemicals and allied products (96)	0.68	Intel	1.66
Radio, TV broadcast stations (25)	1.37	Merck	0.40
General merchandise stores (26)	1.02	Microsoft	1.74
Food stores (26)	0.55	Pepsico	0.68
Computer programming, data processing (11)	1.73	Wal-Mart Stores	0.98

* Numbers in parentheses are firms in sample.

Investment Pyramid







BONDS

THE WALL STREET JOURNAL.

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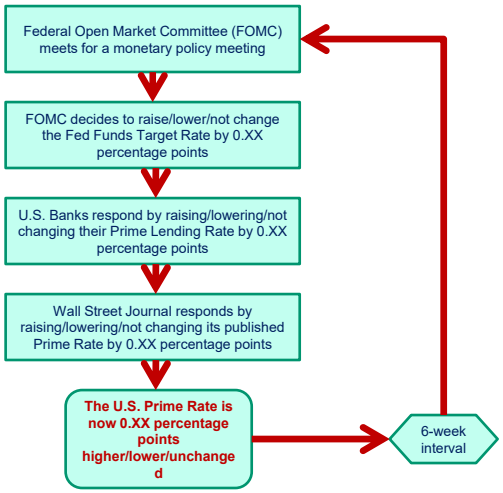
Consumer Rates 9/03/21

	YIELD/RATE %		52 WEEK		CHG IN PCT PTS	
	LAST	WK AGO	HIGH	LOW	52-WK	3-YR
Federal-Funds Rate Target	0.00 - 0.25	0.00 - 0.25	--	--	0.00	-1.75
WSJ Prime Rate*	3.25	3.25	3.25	3.25	0.00	-1.75
Money Market, Annual Yield	0.08	0.08	0.23	0.08	-0.15	-0.39
Five-Year CD, Annual Yield	0.44	0.44	0.65	0.42	-0.21	-1.42
30-Year Mortgage, Fixed	3.07	3.06	3.37	2.83	-0.01	-1.55
15-Year Mortgage, Fixed	2.34	2.38	2.64	2.28	-0.20	-1.72
Jumbo Mortgage, Fixed	3.10	3.09	3.41	2.85	-0.03	-1.73
Five-Year Adj Mortgage (ARM)	2.87	2.86	3.43	2.83	-0.43	-1.59
New-Car Loan, 48-Month	3.72	3.89	4.23	3.72	-0.45	-0.71

* Base rate posted by at least 70% of the nation's largest banks.
Federal-funds, prime rate updated as needed late evening. All other rates updated by 7PM ET.
Sources: Dow Jones Market Data, Bankrate.com.

Update 4/28/2025
- **Fed Funds Rate Target = 4.50%**
- **U.S. Prime Rate = 7.50%**

U.S. Prime Rate Flow Chart



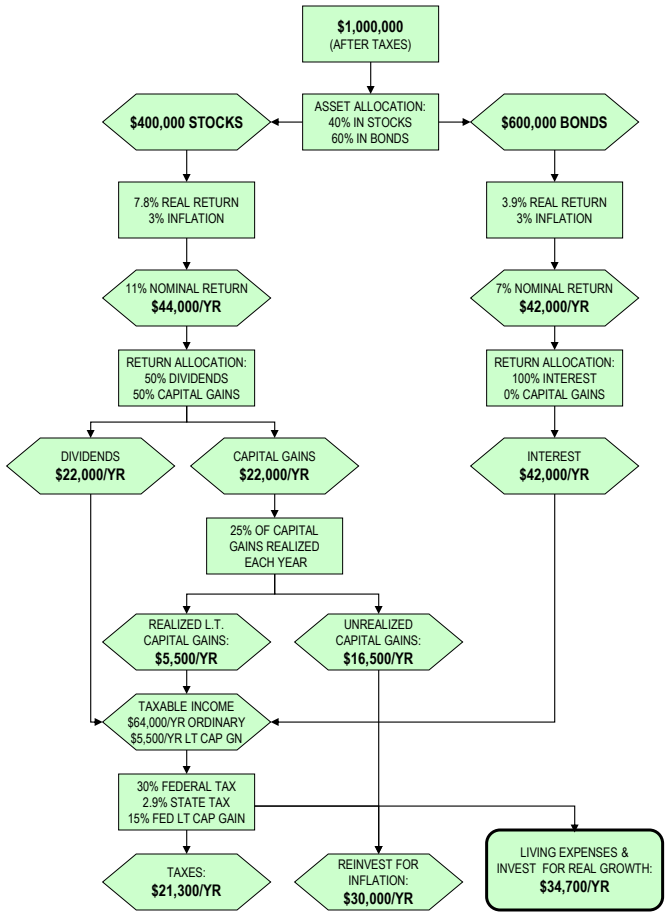
Federal Funds Rate

Updated Apr 28, 2025

Getting to "The Number" …

- Home(s)
  - Mortgage(s) or rent
  - Utilities: gas/elec/water/cable
  - Maintenance
  - Insurance
  - Real estate taxes
- Food
- Clothing
- Entertainment
  - Dining out
  - Sports, shows & cultural events
  - Other fun things
- Vacation travel
- Tuition
- Automobile(s)
  - Payments
  - Fuel
  - Insurance
  - Maintenance
- Gifts & Philanthropy
- Medical
  - Insurance
  - Out-of-pocket costs
- Sundries & Misc.
- Other?

	Per Month
Home (1)	$6,000
Food	800
Clothing	500
Entertainment	700
Vacation Travel	1,500
Tuition & Fees	5,000
Automobiles (2)	3,000
Medical	2,000
Sundries & Misc	200
Subtotal	$19,700
Philanthropy (10%)	1,970
Total per month	$21,670
Total per year	**$260,040**



"The Number" would be …